

EUNWHA PARK

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Summary

Backend Engineer at Samsung Electronics with 3+ years of experience architecting high-traffic distributed systems. Led a **zero-downtime migration of 78B+ rows (50TB+)** across **3 global regions**, eliminating **\$2.2M/year** in database costs. Specialized in Java/Spring, distributed databases (Cassandra, DynamoDB), and cloud-native infrastructure on AWS/EKS.

Professional Experience

Backend Software Engineer

Jan 2023 - Present

Samsung Electronics

Database Migration (ClustrixDB→DynamoDB + Cassandra)

- Led a **zero-downtime migration of 78B+ rows (50TB+)** across **3 global regions**, achieving **100% data integrity** via a hybrid cutover strategy using dual write, shadow read, and per-user flag routing.
- Reduced database operating costs by **95% (\$2M+/year)** by replacing ClustrixDB with a polyglot persistence architecture (DynamoDB + Cassandra).
- Redesigned the legacy RDB schema into a **DynamoDB single-table model**, reducing critical-path DB access from **4 calls to 1** and cutting DB operations by **75%**.

Query Offloading (ClustrixDB→Cassandra)

- Cut **\$700K+/year** in infrastructure costs by migrating write-heavy workloads to Cassandra, reducing RDB write load by **50%** and lock wait times by **68%** globally at **18k+ QPS**.

IAM Modernization

- Redesigned a **10-year legacy IAM monolith** into a **DDD-based modular architecture**, introducing a multi-tier UID system to reduce potential PII exposure and enable decoupling of **10+ downstream services**.

Batch Platform & EKS Migration

- Right-sized thread pools for a **56M+** user batch platform, cutting runtime 61% and eliminating OOM errors.
- Built concurrent batch pipelines with Reactor Kafka and integrated with InfluxDB + Grafana for operational visibility.
- Led multi-region **VM-to-EKS migration**, cutting incident **MTTR** from minutes to seconds; implemented S3-based coordination to eliminate **DynamoDB hot-partition** risks.

Open-Source Contributions

Apache Zeppelin

2025

- Contributed to Apache Zeppelin open-source data notebook platform: 8 PRs merged upstream including NPE prevention and interpreter maintainability improvements.

Kubernetes Website

- Localized core Kubernetes documentation to Korean to enhance accessibility for global developers.

Technical Skills

Languages: Java, Kotlin, Python

Backend & Data: Spring Boot, Cassandra, Kafka, DynamoDB, S3, MySQL, PostgreSQL

Cloud & DevOps: AWS (Certified Solutions Architect – Professional), Kubernetes, Docker, Jenkins, GitHub Actions

Education

University of Illinois Urbana-Champaign | Master of Computer Science (MCS)

Fall 2026

Expected May 2028

Kyungpook National University | B.S. in Mobile Engineering, GPA: 4.45/4.50

Mar 2019 - Feb 2023

- Exchange Student at Lakehead University and Yale University Summer Session.

CERTIFICATIONS

AWS Certified Solutions Architect - Professional